Hongda Su

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Education

University of Science and Technology of China Sep. 2023 – Present
Department of Thermal Science and Energy Engineering

- **Major:** Energy and Power Engineering
- **Minor:** Chemistry (**Dual Degree Track**)
- **GPA: 3.77/4.30** (Major Ranking **1/47**)

Publication

- X. Cao#, Y. Liu#, B. Zhang#, **H. Su#**, et al., Programmable Electrothermal Quad-Functional Metamaterials for Decoupled Multi-Field Control, ***Advanced Materials***, DOI: 10.1002/adma.202522718
- Y. Liu#, J. Li#, Z. Wang#, **H. Su#**, et al., Transient Thermal Camouflage Modulation Enabled by Time-Diffusion Metamaterials (In preparation)
- J. Li, Y. Liu, X. Cao, **H. Su**, et al., Decoupling Conjugate Fields in Diffusion Systems via Field-Line Coordinate Transformations, ***Small Structures***, DOI: 10.1002/sstr.70642
- Z. Tong#, Y. Liu#, J. Li#, Y. Luo#, Z. Wang, Z. Zhang, **H. Su**, et al., Inverse Design of Multisource Thermal Metamaterials for Multifunctional Thermal Camouflage, ***Advanced Functional Materials***, DOI: 10.1002/adfm.78866

#: co-first author

Honors and Awards

- National Natural Science Foundation of China (**NSFC**) — **Youth Student Basic Research Project** Sep. 2026
  National-level research grant, 150 recipients per year
- Bambu Lab Scholarship (top 5%) Oct. 2026
- Wu Zhonghua Talent Program Scholarship (top 20%) Nov. 2025
- CAS Guangzhou Institute of Energy Conversion Scholarship (top 20%) Oct. 2024
- Outstanding Freshman Scholarship (top 50%) Dec. 2023

Research Experience

Design of a Hydrogel-based Moisture-Electric-Generator (MEG) Jul. 2026 – Oct. 2026
with Prof. Will Chang Liu (Department of Mechanical Engineering, NUS, Singapore)

- Preparing the transition-metal-hydroxide electrodes with electrochemical deposition and further characterizing them.
- Fabricating the device and testing its performance.

Design of a Thermal Metamaterial for Transient Thermal Camouflage Oct. 2025 – Present
with Prof. Liqun He (School of Engineering Science, USTC, China)

- Proposing a method of designing thermal metamaterials to reshape heat sources during transient heat conduction.
- Developing 2D and 3D models with script-assisted COMSOL Multiphysics.
- Conducting the FEA simulations.

Design of an Electrothermal Metamaterials for Decoupling Potential and Flux Control May. 2025 – Dec. 2025
with Prof. Liqun He (School of Engineering Science, USTC, China) and Prof. Gang Zhao (School of Information Science and Technology, USTC, China)

- Assisting in developing 2D and 3D Metamaterials models.
- Conducting FEA simulations with Comsol.
- Experimentally verifying the Metamaterials' performance under thermal and electric fields.

Skills

- **English Tests:** TOEFL 99 (MyBest 101), GRE 320 (Q: 168, V: 152)
- **Programming:** C/C++, Matlab, LaTeX, Vibe Coding.
- **Software:** Solidworks, COMSOL, Auto CAD, ChemDraw, Jade, ImageJ, Blender.